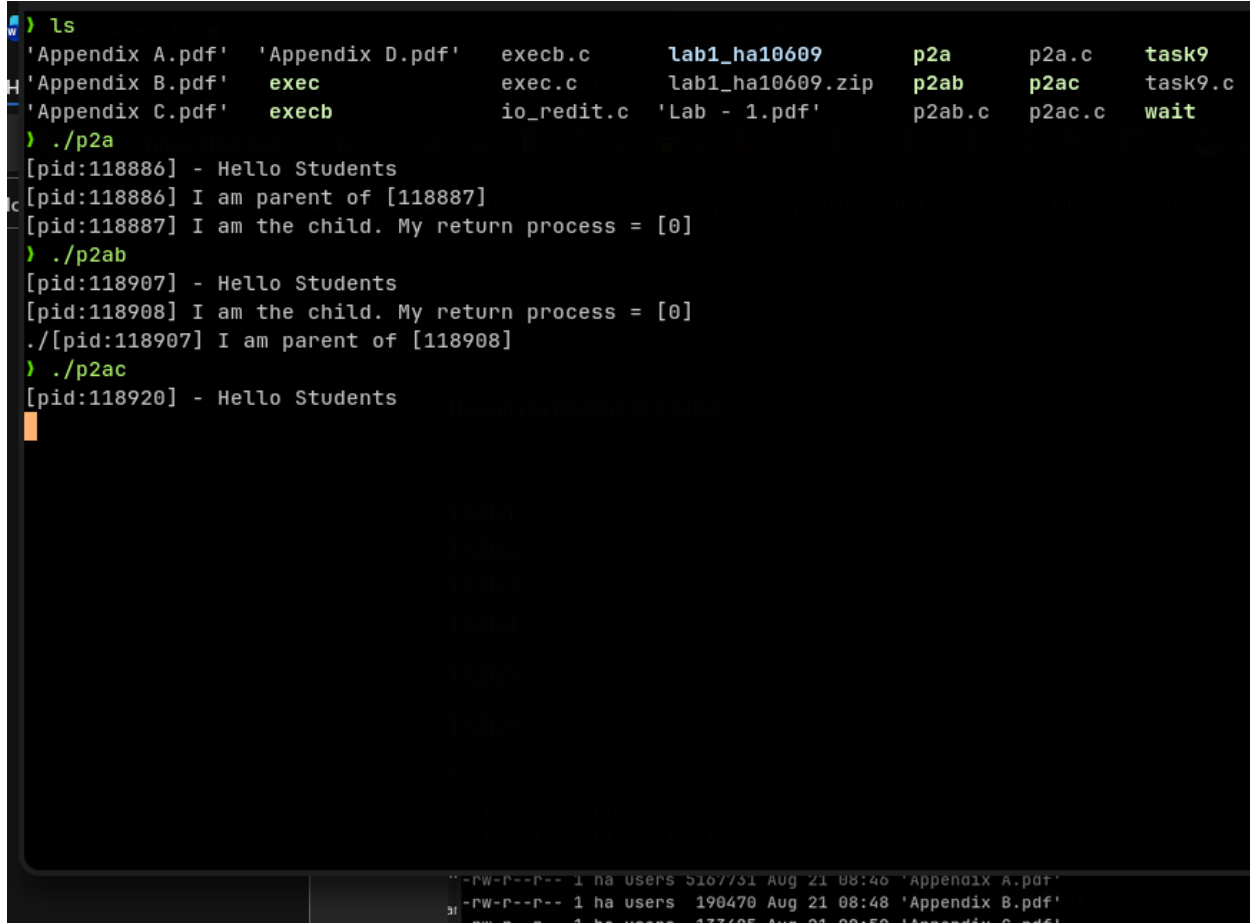


Hassan (ha10609) OS LAB-2

TASK1, TASK2, TASK3:

```
> ls
'Appendix A.pdf'  'Appendix D.pdf'  execb.c          lab1_ha10609      p2a              p2a.c            task9
'Appendix B.pdf'  exec               exec.c           lab1_ha10609.zip  p2ab             p2ac             task9.c
'Appendix C.pdf'  execb             io_reddit.c     'Lab - 1.pdf'     p2ab.c           p2ac.c           wait
> ./p2a
[pid:118886] - Hello Students
[pid:118886] I am parent of [118887]
[pid:118887] I am the child. My return process = [0]
> ./p2ab
[pid:118907] - Hello Students
[pid:118908] I am the child. My return process = [0]
./[pid:118907] I am parent of [118908]
> ./p2ac
[pid:118920] - Hello Students
```

A terminal window with a dark background. The main area shows the execution of three programs: p2a, p2ab, and p2ac. Each program prints a "Hello Students" message and then prints a message indicating its parent process ID. p2a's parent is [118886], p2ab's parent is [118907], and p2ac's parent is [118907]. At the bottom of the terminal, there is a horizontal bar containing a file listing for 'Appendix A.pdf', 'Appendix B.pdf', and 'Appendix C.pdf'.

TASK4, TASK5, TASK6:

```
> ./wait
Hello world (pid:119130)
hello, I am child (pid:119131)
hello, I am parent of 119131 (wc:119131) (pid:119130)
> ./waitb
hello, I am child 1 (pid:119144)
hello, I am parent of 119144 (wc:119144) (pid:119143)
hello, I am child 2 (pid:119145)
hello, I am parent of 119145 (wc:119145) (pid:119143)
hello, I am child 3 (pid:119146)
hello, I am parent of 119146 (wc:119146) (pid:119143)
> ./waitc
hello, I am child 1 (pid:119159)
hello, I am child 2 (pid:119160)
Parent of child 1 (wc:119159) (pid:119158)
Parent of child 2 (wc:119160) (pid:119158)
hello, I am child 3 (pid:119161)
Parent of child 3 (wc:119161) (pid:119158)
```

22 Part I -- Getting Started

Extension: Instead of exiting immediately, have the children perform a small task (like a brief loop or printing different messages multiple times) to make the wait more meaningful.

Let us examine wait.c below:

TASK7:

TASK7, 8:

```
> gcc exec.c -o exec
> ./exec
Hello world (pid:115765)
hello, I am child (pid:115766)
total 9980
-rw-r--r-- 1 ha users 5167731 Aug 21 08:46 'Appendix A.pdf'
-rw-r--r-- 1 ha users 190470 Aug 21 08:48 'Appendix B.pdf'
-rw-r--r-- 1 ha users 133485 Aug 21 08:50 'Appendix C.pdf'
-rw-r--r-- 1 ha users 162413 Aug 21 08:53 'Appendix D.pdf'
-rwxr-xr-x 1 ha users 16424 Aug 28 10:42 exec
-rw-r--r-- 1 ha users 1104 Aug 28 10:40 exec.c
drwxr-xr-x 2 ha users 4096 Aug 21 11:13 lab1_ha10609
-rw-r--r-- 1 ha users 53546 Aug 21 11:14 lab1_ha10609.zip
-rw-r--r-- 1 ha users 4375709 Aug 21 08:40 'Lab - 1.pdf'
-rwxr-xr-x 1 ha users 16208 Aug 28 09:07 p2a
-rwxr-xr-x 1 ha users 16264 Aug 28 09:07 p2ab
-rw-r--r-- 1 ha users 538 Aug 28 09:07 p2ab.c
-rw-r--r-- 1 ha users 511 Aug 28 09:07 p2a.c
-rwxr-xr-x 1 ha users 16264 Aug 28 09:13 p2ac
-rw-r--r-- 1 ha users 541 Aug 28 09:13 p2ac.c
-rwxr-xr-x 1 ha users 16264 Aug 28 09:33 wait
-rw-r--r-- 1 ha users 1621 Aug 28 10:10 waitb.c
-rw-r--r-- 1 ha users 741 Aug 28 09:29 wait.c
-rw-r--r-- 1 ha users 1581 Aug 28 10:26 waitc.c
hello, I am parent of 115766 (wc:115766) (pid:115765)
> touch exeb.c
> nvim
```

TASK9:

```

) nvim .
) ./task9 "abra cadabra" io_redit.c
hello, I am child 1 (pid:118744)
hello, I am child 3 (pid:118746)
    fprintf(stderr, "fork failed\n");
hello, I am child 2 (pid:118745)
hello, I am child 4 (pid:118747)
Parent of child 1 (wc:118744) (pid:118743)
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <sys/stat.h>

int main(int argc, char *argv[]) {
    int rc = fork();

    if (rc < 0) {
        // fork failed
        fprintf(stderr, "fork failed\n");
        exit(1);
    }
    else if (rc == 0) {
        // child: redirect standard output to a file
        close(STDOUT_FILENO);
        open("newfile.txt", O_CREAT | O_WRONLY | O_TRUNC, S_IRWXU);

        char *myargs[3];

```

```
fprintf(stderr, "fork failed\n");
exit(1);
}
else if (rc == 0) {
    // child: redirect standard output to a file
    close(STDOUT_FILENO);
    open("newfile.txt", O_CREAT | O_WRONLY | O_TRUNC, S_IRWXU);

    char *myargs[3];
    myargs[0] = strdup("wc"); // program: wc (word count)
    myargs[1] = strdup("p3.c"); // arg: file to count
    myargs[2] = NULL; // mark end of array

    execvp(myargs[0], myargs); // runs word count
}
else {
    // parent goes down this path (main)
    int rc_wait = wait(NULL);
}

return 0;
}
35 109 861 io_redit.c
abra cadabra
Parent of child 2 (wc:118745) (pid:118743)
Parent of child 3 (wc:118746) (pid:118743)
Parent of child 4 (wc:118747) (pid:118743)
```

~/Documents/Notebooks/OS/LAB  main ?17